Model Selection and Regularization

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Linear regression

Describes the relationship between an outcome variable Y and a set of predictor variables Xs:

$$Y = \beta_o + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon, \quad \epsilon \sim N(0, \sigma^2)$$

- This model is usually fitted by using least squares.

Minimizing the residual sum of squares:

$$\text{RSS} = \sum_{i=1}^{n} (y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij})^2$$

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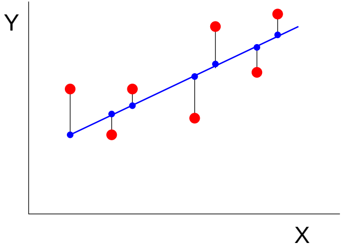


Prediction accuracy and interpretability

Alternative fittings can be better.

- Prediction accuracy
  - Bias: if the relations are approximately linear, least squares have low bias.
  - Variance:
    - if $n \gg p$ least squares estimates have low variance;
    - if n not much larger than p least squares estimates can have large variance, leading to overfitting and poor predictions.
  - Methods constraining or shrinking the estimates can perform much better.
- Interpretability
  - Several Xs might not be associated with Y and just add to the complexity of the model.
  - Methods for automatic variable selection can lead to much more interpretable models.

Some alternatives

- Subset selection:
  - Identify a subset of predictor variables.
  - Use least squares on the reduced set of variables.
- Shrinkage or regularization: fitting a model with all p predictors but shrinking the estimates to 0.
  - Reduces variance of the estimates.
  - Can select variables by shrinking some estimates to 0.
- Dimension reduction: creating new (fewer) predictors from the original ones.
  - Project the original p predictors to a subspace with dimensions m, with $m < p$.
  - Use least squares on a model with the new variables.

Best subset selection

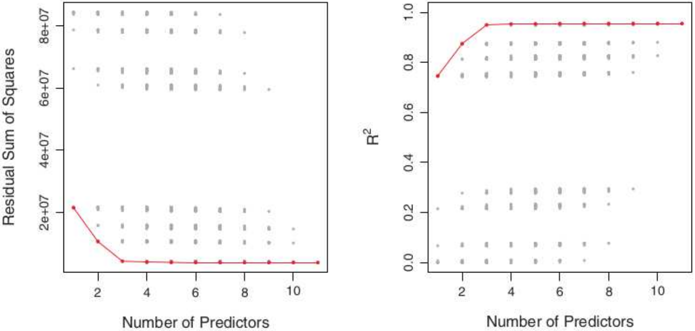
Algorithm:

1. Define the null model $\mathcal{M}_0$ with no predictors (predicts the sample mean for all observations).
2. For $k = 1, 2, \ldots, p$:
   a. Fit all $\binom{p}{k}$ models that contain k predictors.
   b. Pick the best (smallest RSS or largest R^2) of these models and call it $\mathcal{M}_k$.
3. Select the best model $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using for example cross-validated prediction error.

- Infeasible for large p. More than one million models for $p = 20$.
- Low RSS and high R^2 indicate low training error but not low **test error**.
- Large search space can lead to overfitting and high variance of estimates.
- The number of samples n has to be larger than number of variables p.

Best subset selection

Example linear regression with $p = 11$ possible predictors:



red: best model for each p

Figure from James et al.

- Performance on the **training data** gets better with p.
- RSS and R^2 cannot be used to choose models across p values.

Stepwise Selection

Methods to search among a smaller set of models.

Algorithm: forward stepwise selection

1. Define the null model $\mathcal{M}_0$ with no predictors.
2. For $k = 0, 1, \ldots, p-1$:
   a. Consider all the $p - k$ models that augment the predictors in $\mathcal{M}_k$ with one.
   b. Pick the best (smallest RSS or largest R^2) of these $p - k$ models and call it $\mathcal{M}_{k+1}$.
3. Select the best model $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using for example cross-validated prediction error.

- Much fewer models to fit. 211 models for $p = 20$.
- RSS and R^2 cannot be used to choose models across p values.
- Alternatively, backward stepwise selection, starting from the full model. Requires $p < n$.

Choosing the optimal model

- Indirectly, adjusting the training error to account for overfitting. Adding a penalty for predictors.
  - C_p, Akaike information criterion (AIC), Bayesian information criterion (BIC) - small for low test error
  - adjusted R^2 - large for low test error

$$C_p = \frac{1}{n}(\text{RSS} + 2d\hat{\sigma}^2) \qquad \text{AIC} = \frac{1}{n\hat{\sigma}^2}(\text{RSS} + 2d\hat{\sigma}^2)$$

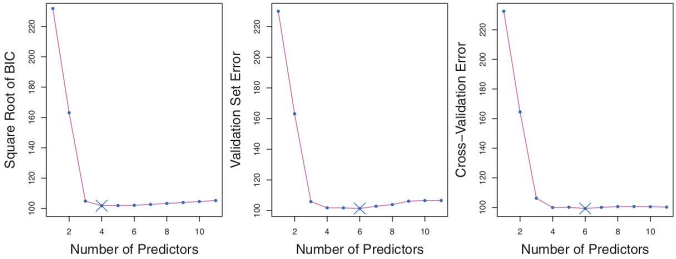
$$\text{BIC} = \frac{1}{n\hat{\sigma}^2}(\text{RSS} + \log(n)d\hat{\sigma}^2) \qquad \text{adjusted}R^2 = 1 - \frac{\text{RSS}/(n-d-1)}{\text{TSS}/(n-1)}$$

$\hat{\sigma}^2$ is an estimate of the variance of the error ϵ
d is the number of predictors
TSS is the total sum of squares of the response $\sum(y_i - \overline{y})^2$

Choosing the optimal model

- Directly, with cross-validation or a validation set.

Example linear regression with 11 possible predictors:



validation set: training on three-quarters of the data

10-fold cross-validation

- One-standard-error rule: If a set of models have very similar test errors, select the smallest model for which the estimated test error is within 1 standard error of the lowest point.

Shrinkage: Ridge regression

The least squares procedure estimates the parameters by minimizing the residual sum of squares:

$$\mathrm{RSS} = \sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij})^2$$

Ridge regression minimizes the same function plus a shrinkage penalty:

$$\sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij})^2 + \lambda\sum_{j=1}^{p}\beta_j^2$$

with tuning parameter $\lambda \geq 0$.

Or in another formulation, minimizing RSS subject to:

$$\sum_{j=1}^{p}\beta_j^2 \leq s$$

Ridge regression

- $\lambda = 0$ least square estimates
- $\lambda \to \infty$ all estimates approach 0
- An appropriate λ has to be set!
  - Cross-validation

Note: use standardized predictors.

$$\tilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2}}$$

Example:

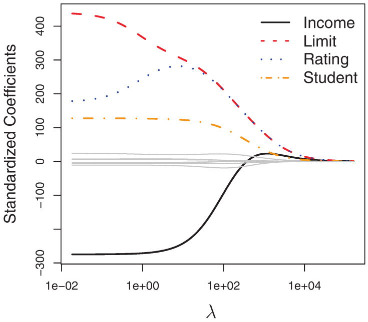


Figure adapted from James et al.

Ridge regression

- Prediction accuracy:
  - RR results in increased bias but decreased variance.
  - RR works well when least squares estimates have high variance.

Simulation, $n = 50, p = 45$

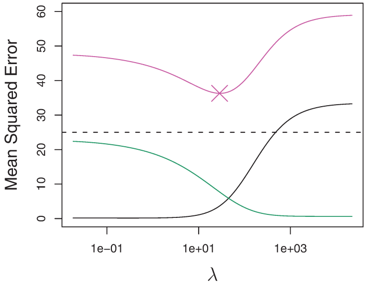
Black: squared bias

Green: variance

Pink: test mean squared error

Dashed: minimum possible MSE

Figure adapted from James et al.



- Interpretability:
  - RR indicates important variables and is computationally faster than variable selection.
  - However the model always keeps all the p variables!

Lasso

Lasso: least absolute shrinkage and selection operator

- Lasso is similar to RR, but it does variable selection.
  - Some parameters will be shrunk to 0.

Lasso minimizes RSS plus a different shrinkage penalty:

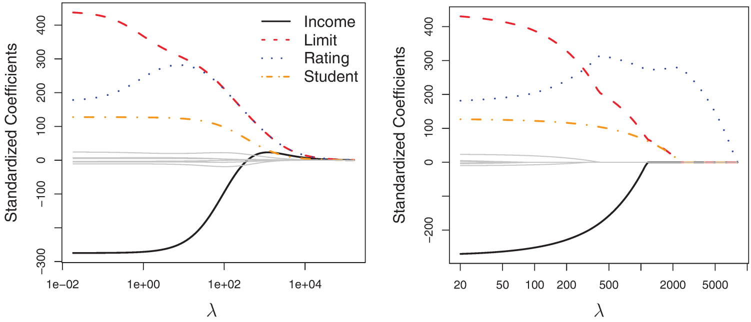
$$\sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij})^2 + \lambda \sum_{j=1}^{p}|\beta_j|$$

with tuning parameter $\lambda \geq 0$.

Or in another formulation, minimizing RSS subject to:

$$\sum_{j=1}^{p}|\beta_j| \leq s$$

Ridge regression versus Lasso



Prediction accuracy:

- Lasso is expected to be better when a small number of predictors have large effects and the rest has almost no effect.
- Ridge regression is expected to be better when many predictors are all weakly related to the outcome variable.

Dimension reduction methods

Idea:

- Transform the original p predictors into $M < p$ new predictors.
- Fit a model to the new predictors.

The new variables $Z_1, Z_2, \ldots, Z_M$ are linear combinations of the p original ones:

$$Z_m = \sum_{j=1}^{p} \phi_{jm} X_j$$

The new model:

$$y_i = \theta_0 + \sum_{m=1}^{M} \theta_m z_{im} + \epsilon_i \qquad i = 1, \ldots, n$$

- When p is large (in relation to n), selecting $M \ll p$ can significantly reduce the variance of the estimates.

Principal component analysis

Idea: find a lower representation of the data that captures as much information as possible.

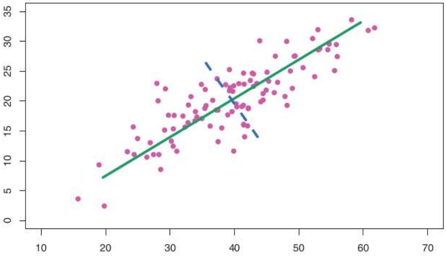
- Find a set of dimensions that are as interesting as possible.
- Interesting: a large variation of the data is on that dimension.

- PCA is also a technique for unsupervised learning .

Principal components

Illustration in 2 dimensions:

- 1st PC - direction through the data that explains the most variance
- 2nd PC - direction orthogonal to that of the 1st component, explaining the next greatest amount of variance



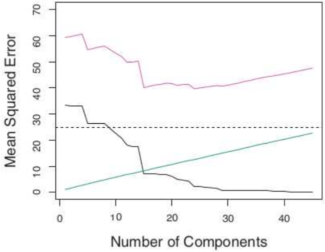
First principal component:

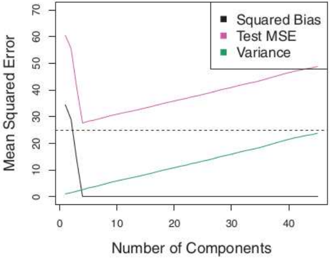
$$Z_1 = \phi_{11} X_1 + \phi_{21} X_2 + \cdots + \phi_{p1} X_p, \quad \text{with} \quad \sum_{j=1}^{p} \phi_{j1}^2 = 1$$

ϕs are the loadings of the first principal component.

Principal component regression

- It assumes that the directions with most variance are the ones associated with Y.
- If the assumption holds and $M \ll p$ overfitting can be reduced.
- Number of components is chosen by cross-validation.
- Predictors are standardized prior to the PCA





Simulated data

Left: Data is a function of all the 45 predictors

Right: Data is a function of the first 5 PCs

Dashed: irreducible error

Visualizing the data

- 50 observations of 4 variables
- Blue dots: Scores of the observations in the first two components.
- Orange arrows: The first two principal components vectors of loadings.

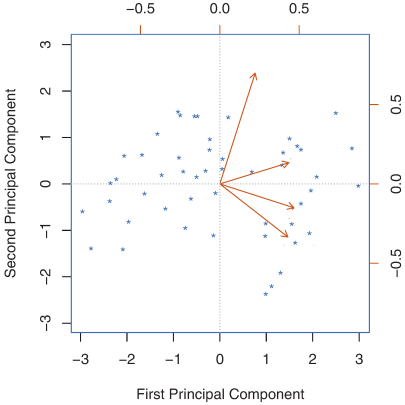


Figure adapted from Jones et al. 2013